

Faults in Capability Space, Across the Fleet

Study I-F — 9 actuator faults applied to every spacecraft in the fleet, and what the spacecraft classes have in common

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1 What this study does

Study I put twelve faults through **one** vehicle’s actuator set — the Apollo LM as the fault-injection studies model it — and found that a fault’s effect on capability is a poor predictor of whether it is survivable. This study asks the prior question: **is what we saw a property of that vehicle, or of the faults?**

The same capability measures are applied to every spacecraft in the actuation-envelope fleet — 18 vehicles with usable actuator geometry, across 5 classes — and then compared between vehicles and between classes.

1.1 Restating the faults so any vehicle can suffer them

The Study H catalogue is written against one plant, so each fault is restated at actuator level:

Fault	Tier	Applied to
Engine out	engine	one unit of that tier, swept over every unit
Thrust reduction, $\eta=0.15$	engine	one unit of that tier, swept over every unit
Thrust reduction, $\eta=0.50$	engine	one unit of that tier, swept over every unit
Thrust excess, $\eta=1.30$	engine	one unit of that tier, swept over every unit
TVC misalignment (3° , 2°)	engine	one unit of that tier, swept over every unit
Gimbal seizure	engine	one unit of that tier, swept over every unit
Valve stuck open (30 % floor)	engine	one unit of that tier, swept over every unit
RCS thruster dead	rcs	one unit of that tier, swept over every unit

Fault	Tier	Applied to
RCS thruster stuck open	rcs	one unit of that tier, swept over every unit

Two conventions are worth stating, because they are choices rather than readings:

- **valve stuck open is 30% of the affected unit’s own rated thrust.** Study I quoted the floor as a multiple of the LM’s hover share, and most of this fleet never hovers. A fraction of the unit’s own rating keeps the structural content of the fault — a floor it cannot throttle below — without borrowing an operating point only a lander has.
- **Worst case, not first case.** Removing “thruster 0” measures the layout’s numbering, not its redundancy. Every single-unit fault is applied to each unit in turn and reported at its worst, which is the number a redundancy argument actually needs.

The faults left out are left out for a reason established in Study I: transport delay, slow thrust response and erosion drift at onset **do not change a static capability set at all**. They are not weak effects here, they are exactly zero, and they were the least survivable faults in Study H.

1.2 Why the fleet is measured by reach and not by volume

Volume is the natural size of a 3-D set and the wrong measure for a degenerate one. A vehicle whose only actuator is a single centreline gimballed engine — Ariane 5, Vega, Vega-C, Juno — has **no roll authority at all**, so its moment set is a flat disc with exactly zero volume. Every retention ratio would be 0/0, and those vehicles would drop out of the comparison entirely: not because they lack authority, but because their authority is planar.

The fleet map is therefore drawn from the **mean support** of each set — its average reach over all directions — which is defined in any dimension, positive whenever the set is, and comparable across vehicles four orders of magnitude apart once normalised by their own healthy value. Volumes and the set’s actual dimension are both kept in `IF_fleet.csv`, and a set that has *lost* a dimension is reported as 0 rather than as a ratio between incommensurable quantities.

2 The fleet map

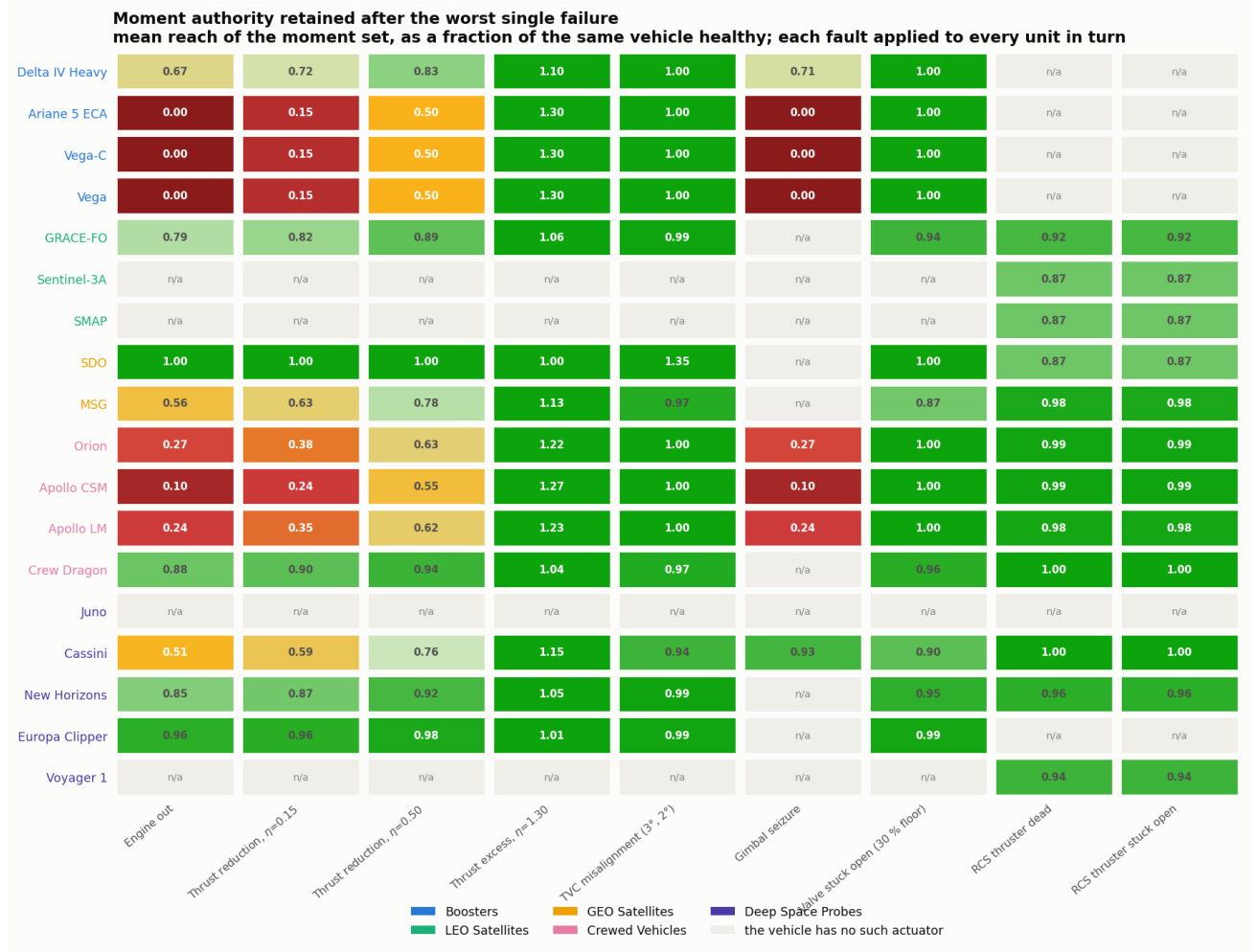


Figure 1: Moment authority retained after the worst single failure

Read the blanks first. They are not missing data — they are vehicles that **cannot suffer that fault**, because they have no actuator of that tier. A spacecraft with no main engine cannot lose one; a spacecraft with no RCS tier cannot lose a thruster. Fault exposure is a function of architecture before it is a function of reliability, and the map shows several vehicles that are immune to entire fault classes by construction.

Across the fleet:

Fault	Vehicles exposed	Mean retained	Range	Lost zero-moment
Gimbal seizure	8	0.28×	0.00× – 0.93×	0
Engine out	14	0.49×	0.00× – 1.00×	0
Thrust reduction, $\eta=0.15$	14	0.56×	0.15× – 1.00×	0
Thrust reduction, $\eta=0.50$	14	0.74×	0.50× – 1.00×	0
RCS thruster stuck open	12	0.95×	0.87× – 1.00×	0
RCS thruster dead	12	0.95×	0.87× – 1.00×	0
Valve stuck open (30 % floor)	14	0.97×	0.87× – 1.00×	0
TVC misalignment ($3^\circ, 2^\circ$)	14	1.01×	0.94× – 1.35×	0
Thrust excess, $\eta=1.30$	14	1.15×	1.00× – 1.30×	0

Gimbal seizure is the most damaging fault on average ($0.28\times$ of healthy moment volume), and **Engine out** is the most variable — its effect ranges from $0.00\times$ to $1.00\times$ depending entirely on which vehicle suffers it. That spread is the study’s first real finding: *the same fault is not the same event on different architectures*.

3 Do the spacecraft classes respond alike?

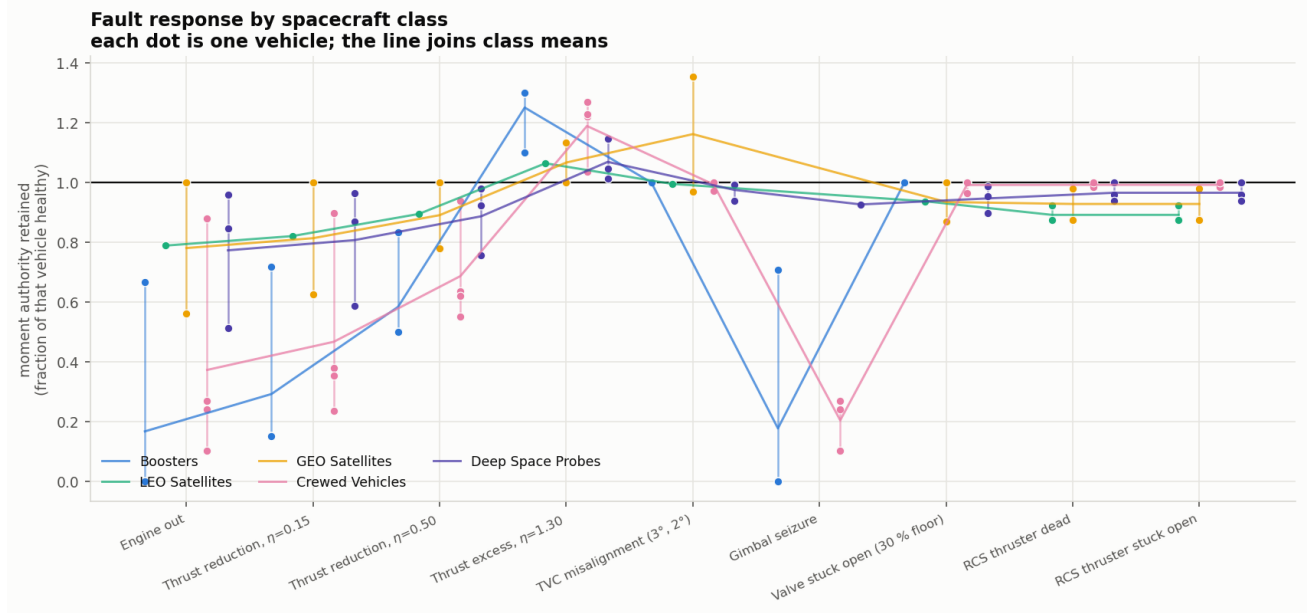


Figure 2: Fault response by spacecraft class

Class	Engine out	Thrust reduction	Thrust reduction	Thrust excess	TVC misalignment	Gimbal seizure	Valve stuck open	RCS thruster dead	RCS thruster stuck open
Boosters	$0.17\times$	$0.29\times$	$0.58\times$	$1.25\times$	$1.00\times$	$0.18\times$	$1.00\times$	—	—
LEO Satellites	$0.79\times$	$0.82\times$	$0.89\times$	$1.06\times$	$0.99\times$	—	$0.94\times$	$0.89\times$	$0.89\times$
GEO Satellites	$0.78\times$	$0.81\times$	$0.89\times$	$1.07\times$	$1.16\times$	—	$0.93\times$	$0.93\times$	$0.93\times$
Crewed Vehicles	$0.37\times$	$0.47\times$	$0.69\times$	$1.19\times$	$0.99\times$	$0.20\times$	$0.99\times$	$0.99\times$	$0.99\times$
Deep Space Probes	$0.77\times$	$0.81\times$	$0.89\times$	$1.07\times$	$0.97\times$	$0.93\times$	$0.95\times$	$0.96\times$	$0.96\times$

3.1 The quantitative answer

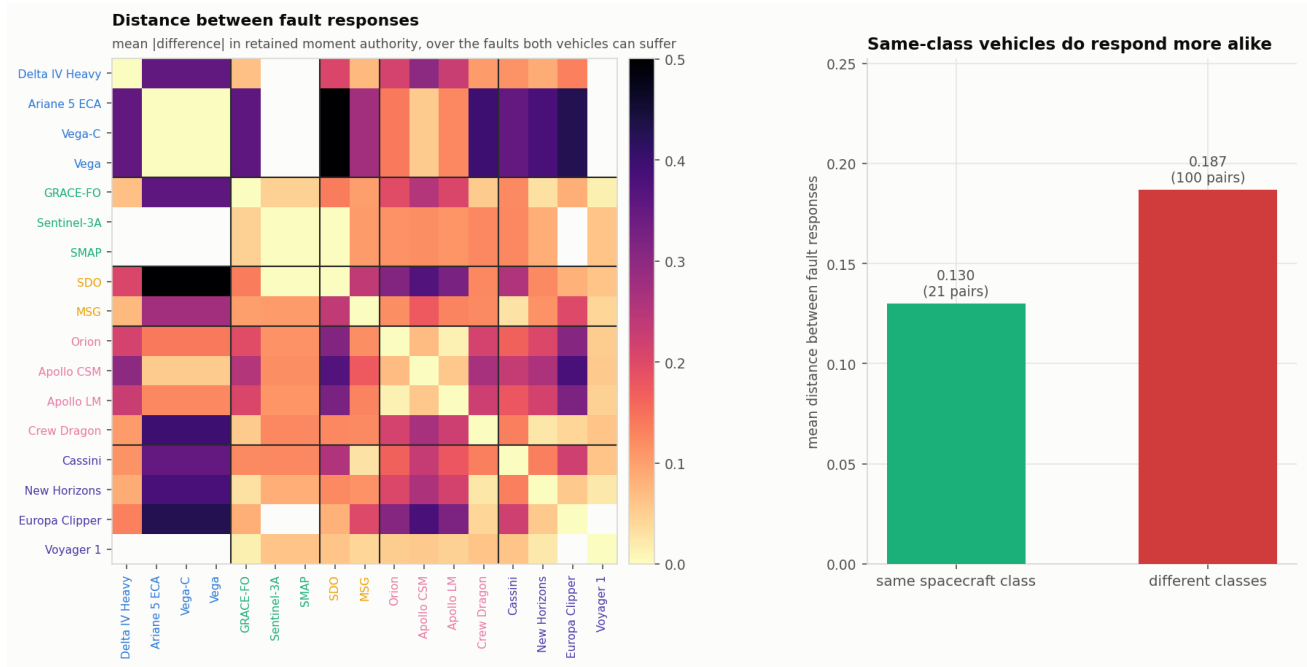


Figure 3: Distance between fault responses, and class membership

The distance between two vehicles is the mean absolute difference in retained moment authority over the faults both can suffer — a direct measure of “do these two react alike”. Averaged over every pair:

Pairing	Mean distance	Pairs
Same spacecraft class	0.130	21
Different classes	0.187	100

Class	Vehicles compared	Mean distance within the class
Boosters	4	0.177
LEO Satellites	3	0.032
GEO Satellites	2	0.237
Crewed Vehicles	4	0.140
Deep Space Probes	4	0.099

The classes are not equally coherent. LEO Satellites, Deep Space Probes hold together more tightly than the fleet average, while GEO Satellites is more spread out than two vehicles picked at random.

Same-class vehicles do respond more alike — 0.130 against 0.187, a gap of 0.057. Mission class is a partial proxy for actuator architecture: vehicles built for the same job tend to be built with the same tiers, and it is the tiers that decide the response.

4 What actually predicts the response

Three architectural facts explain most of the map, and none of them is the spacecraft’s mission.

1. Tier count decides exposure. 8 vehicles carry a single main engine, so **engine out** is not a degradation for them — it is the total loss of the tier, and the moment set collapses to whatever the RCS ring can produce,

or to nothing if there is none. Those are: Apollo CSM, Apollo LM, Ariane 5 ECA, Juno, Orion, SDO, Vega, Vega-C.

By contrast Crew Dragon, Delta IV Heavy, Europa Clipper, New Horizons carry three or more, and lose a fraction rather than a tier.

2. A thruster ring is a different kind of redundancy from an engine cluster. Study I found this on one vehicle — sixteen thrusters near-interchangeable, two engines not — and the fleet confirms it as a rule: RCS-tier faults are consistently the mildest column of the map, and engine-tier faults the harshest, regardless of class. A ring of many small units degrades gracefully; a cluster of few large ones degrades in steps.

3. Gimbals convert thrust into moment, so seizing one costs only vehicles that have them. gimbal seizure is n/a for every vehicle whose engines are rigidly mounted — and for those, the entire moment tier is the RCS ring, which is why their RCS thruster dead numbers matter more than anyone else’s.

5 What carries over from the single-vehicle study, and what does not

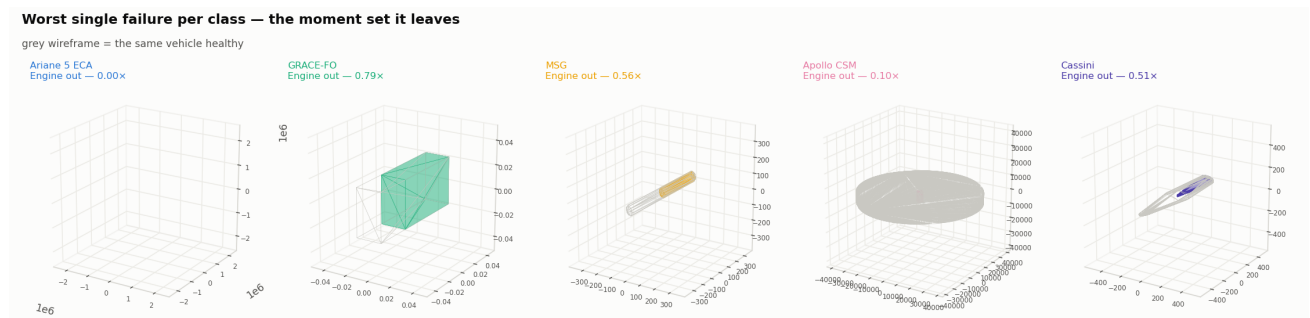


Figure 4: The worst single failure per class

Carries over. The four effect categories — shrink, enlarge, deform, puncture — describe every vehicle in the fleet, not just the LM. A stuck-open valve punctures the set on every vehicle it can strike, taking the origin with it while leaving most of the volume intact. Thrust excess enlarges the set asymmetrically everywhere. The taxonomy is a property of the faults.

Does not carry over. The *magnitudes* are not transferable at all. Engine out costs one vehicle 0.00x and another 1.00x. Any number quoted from a single-vehicle capability study — including Study I’s — is a number about that vehicle’s actuator layout, and the honest way to use it is as a method rather than a result.

And the central negative result is architectural, not vehicle-specific. Every fault that changed nothing in Study I’s capability space changes nothing here either, on any vehicle in the fleet, because a transport delay is not a statement about actuator geometry at all. Capability space remains necessary and insufficient across the whole fleet: it tells you what a damaged vehicle can do at an instant, and nothing about whether it can do it in time.

6 Limitations

- **Geometry is largely estimated.** The workbook supplies thrust magnitudes and counts; positions, directions, gimbal limits and inertias come from `vehicles.py`, and only the Apollo LM’s layout is this repository’s own validated model. The comparisons are therefore between *models* of these spacecraft.
- **Instantaneous sets, again.** No actuator dynamics, no rate limits, no detection. The temporal fault classes are invisible here by construction.
- **Worst-case single failures only.** Simultaneous faults, and the fault-on-fault cases a long mission actually accumulates, are not covered.

- **One operating point is missing.** Study I could condition its moment sets on hover thrust because the LM hovers. There is no fleet-wide equivalent, so the fleet numbers are unconditional set volumes — the more flattering measure, as Study I showed.
- **0 vehicles have no usable actuator geometry** and are excluded rather than counted as zero.
- **Juno had to be left out of the similarity comparison entirely**, for having zero moment authority in the model: a single centreline engine, rigidly mounted, and no modelled RCS layout. Every retention ratio for such a vehicle is 0/0. That is a gap in the geometry database rather than a spacecraft without attitude control, and it is the same gap the reliability study flags for this vehicle’s thruster tier.

7 Reproducing

```
python studies/capability_space/run_fleet_capability.py    # the fleet sweep
python studies/capability_space/plot_fleet_capability.py  # figures
python studies/capability_space/build_fleet_report.py     # this document
```